

Generalized linear models

Estimation equation for GLMs

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Estimation equation for GLMs - Contents

Estimation of GLMs

Estimation of GLMs

- ▶ Goal: Estimation of a GLM with observations $(y_i, \mathbf{x}_i), i = 1, \dots, n$ with $y_i | \mathbf{x}_i$ conditionally independent !
- ▶ Known structure $(\vartheta, \phi, b(\cdot), c(\cdot, \cdot, \cdot))$
→ to estimate: β
- ▶ Classical method: Maximum Likelihood

$$\eta = \mathbf{x}^T \beta$$

Likelihood

$$L(\beta) = f(y_1, \dots, y_n | \mathbf{x}_1, \dots, \mathbf{x}_n, \beta)$$

cond. ind.

$$= \prod_{i=1}^n f(y_i | \mathbf{x}_i, \beta)$$

Estimation of GLMs

$$y_i = w(h(x_i^T \beta))$$

log-Likelihood

$$l(\beta) = \log L(\beta) = \sum_{i=1}^n \log f(y_i | \mathbf{x}_i, \beta)$$

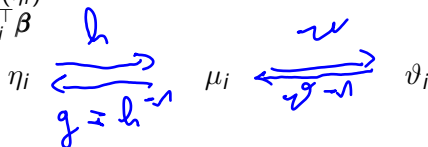
$\underbrace{\hspace{10em}}_{l_i(w_i)}$

$$\beta = (\beta_0, \beta_1, \dots, \beta_p)^T = \sum_{i=1}^n \frac{y_i \vartheta_i - b(\vartheta_i)}{\phi} + c(y_i, \phi)$$

► Connection to ϑ :

- $\vartheta_i = \vartheta(\mu_i)$
- $\mu_i = h(\eta_i)$
- $\eta_i = \mathbf{x}_i^T \beta$

w is a function of μ_i



Estimation of GLMs

canonical link $\eta \rightarrow h^{-1}$
 $\Rightarrow \vartheta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$

Finally:

$$\vartheta_i = \vartheta(h(\mathbf{x}_i^\top \boldsymbol{\beta}))$$

For the score function that means:

$$\mathbf{s}(\boldsymbol{\beta}) = \frac{\partial l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n \frac{\partial l(\vartheta_i)}{\partial \vartheta} \cdot \frac{\partial \vartheta(\mu_i)}{\partial \mu} \cdot \frac{\partial h(\eta_i)}{\partial \eta} \cdot \frac{\partial \eta_i}{\partial \boldsymbol{\beta}}$$

$$\frac{\partial l(\vartheta_i)}{\partial \vartheta} = \frac{y_i - b'(\vartheta_i)}{\phi} = \frac{y_i - \mu_i}{\phi}$$

$$\frac{\partial \vartheta(\mu_i)}{\partial \mu} \stackrel{(*)}{=} \frac{1}{\frac{\partial \mu}{\partial \vartheta}(\vartheta_i)} = \frac{1}{b''(\vartheta_i)} = \frac{\phi}{\text{Var}(y_i)} = \frac{1}{v(\mu_i)}$$

$$\frac{\partial h(\eta_i)}{\partial \eta} = \text{depends on the model}$$

$$\frac{\partial \eta_i}{\partial \boldsymbol{\beta}} = \mathbf{x}_i$$

Ⓟ "inverse rule"

$$f'(x) = \frac{1}{(f^{-1})'(f(x))}$$

variance function

Estimation of GLMs

Score function

$$\begin{aligned}\Rightarrow \mathbf{s}(\beta) &= \sum_{i=1}^n \frac{y_i - \mu_i}{\phi} \cdot \frac{1}{v(\mu_i)} \cdot \frac{\partial h(\eta_i)}{\partial \eta} \cdot \mathbf{x}_i \\ &= \sum_{i=1}^n \frac{\partial h(\eta_i)}{\partial \eta} \cdot \frac{y_i - \mu_i}{\phi v(\mu_i)} \cdot \mathbf{x}_i\end{aligned}$$

Special case: canonical link: $\vartheta_i = \eta_i = \mathbf{x}_i^\top \beta$

$\vartheta_i = \eta_i \xrightarrow{h = \eta \mapsto \mu} \mu_i \xrightarrow{g = \mu \mapsto \vartheta} \vartheta_i = \eta_i$

$$\Rightarrow \mathbf{s}(\beta) = \frac{\partial l(\vartheta_i)}{\partial \vartheta} \cdot \frac{\partial \eta_i}{\partial \beta} = \sum_{i=1}^n \frac{y_i - \mu_i}{\phi} \mathbf{x}_i$$

Particularly, we also get:

$$\Rightarrow \frac{\partial h(\eta_i)}{\partial \eta} = v(\mu_i) = \frac{\text{Var}(y_i)}{\phi}$$

via setting the boxes equal

Estimation of GLMs

Example: Normal distribution

$$\eta(\cdot) = \text{id}(\cdot)$$

$$\frac{\partial \eta(\eta_i)}{\partial \eta_i} = 1, \quad \text{var}(y_i) = \sigma^2, \quad \phi = \sigma^2$$

$$\Rightarrow \underline{\beta} = \sum_{i=1}^n \frac{y_i - \mu_i}{\sigma^2} \underline{x}_i$$

(canonical link)

Estimation of GLMs

Example: Log-linear Poisson model

canonical link, $\phi = 1$

$$h(\eta_i) = \exp(\eta_i) \Rightarrow \frac{\partial h(\eta_i)}{\partial \eta} = \exp(\eta_i)$$

$$\text{Var}(y_i) = \lambda_i = \mathbb{E}(y_i) = \mu_i = h(\eta_i)$$

$$\underline{\ell}(\underline{\beta}) = \sum_{i=1}^n \exp(\eta_i) \frac{y_i - \mu_i}{\exp(\eta_i)} \underline{x}_i = \sum_{i=1}^n (y_i - \mu_i) \underline{x}_i$$

Information matrices

- ▶ Information matrices determine the asymptotic variance of an estimator in ML inference

Fisher information

- ▶ Observed:

(negative
Hessian)

$$\mathbf{F}_{\text{obs}}(\boldsymbol{\beta}) = -\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^2} = \left(-\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \beta_i \partial \beta_j} \right)_{ij}$$

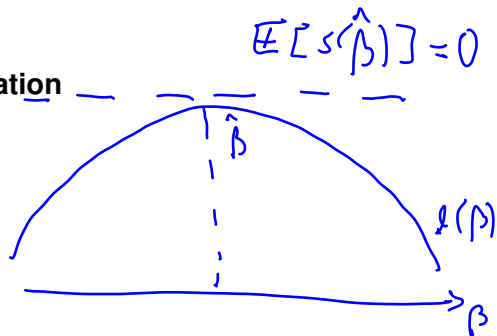
- ▶ Expected

$$\mathbf{F}(\boldsymbol{\beta}) = \mathbb{E}(\mathbf{F}_{\text{obs}}(\boldsymbol{\beta}))$$

- ▶ $\mathbf{F}_{\text{obs}}(\boldsymbol{\beta})$ is a random variable, as it depends on random observations, $\mathbf{F}(\boldsymbol{\beta})$ doesn't

Information matrices

Interpretation Fisher information



$\Rightarrow s(\beta)$ reflects the slope/increase of tangent

$\Rightarrow F(\beta)$ the curvature

$\rightarrow$ large values : strong curvature $\cap$
 $\rightarrow$ small values : weak curvature $\smile$

Information matrices

- One-dimensional (asymptotic):

$$\text{Var}(\hat{\beta}) \approx \left(-\frac{\partial^2 l(\hat{\beta})}{\partial \beta^2} \right)^{-1}$$

- Vectorial:

1. density exists
2. $\theta \in \Theta$ open interval

$$\text{Var}(\hat{\beta}) \approx \left(F_{\text{obs}}(\hat{\beta}) \right)^{-1}$$

- Under Fisher regularity:

$$\textcircled{*} \mathbb{E}(\mathbf{s}(\beta)) = \mathbf{0}$$

can be shown

3. support indep. from β
4. diff. and int. are
and interchangeable

↳ Expo. fam. ✓

$$\mathbb{E} \left(\underbrace{-\frac{\partial^2 l(\beta)}{\partial \beta^2}}_{=v} \right) \downarrow = \mathbb{E} \left(\left(\frac{\partial l(\beta)}{\partial \beta} \right) \left(\frac{\partial l(\beta)}{\partial \beta} \right)^{\top} \right)$$

$$\mathbb{E} \left[\underbrace{\left(\mathbf{s}(\beta) - \mathbb{E}[\mathbf{s}(\beta)] \right) \cdot \left(\mathbf{s}(\beta) - \mathbb{E}[\mathbf{s}(\beta)] \right)^{\top}}_{=v} \right] = \mathbb{E} \left(\mathbf{s}(\beta) \mathbf{s}(\beta)^{\top} \right) \textcircled{*} = \text{Cov}(\mathbf{s}(\beta))$$

Information matrices

Hence:

$$\begin{aligned}\mathbf{F}(\boldsymbol{\beta}) &= \mathbb{E} \left(-\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^2} \right) = \mathbb{E} \left(\mathbf{s}(\boldsymbol{\beta}) \mathbf{s}(\boldsymbol{\beta})^\top \right) \\ &= \mathbb{E} \left(\left(\sum_{i=1}^n \frac{\partial h(\eta_i)}{\partial \eta} \cdot \frac{y_i - \mu_i}{\phi v(\mu_i)} \cdot \mathbf{x}_i \right) \left(\sum_{i=1}^n \frac{\partial h(\eta_i)}{\partial \eta} \cdot \frac{y_i - \mu_i}{\phi v(\mu_i)} \cdot \mathbf{x}_i \right)^\top \right)\end{aligned}$$

The expectation is with respect to the random variable y_i . The product of the sum contains the terms:

$$\left(\frac{\partial h(\eta_i)}{\partial \eta} \right) \left(\frac{\partial h(\eta_j)}{\partial \eta} \right) \frac{(y_i - \mu_i)(y_j - \mu_j)}{\text{Var}(y_i)\text{Var}(y_j)} \mathbf{x}_i \mathbf{x}_j^\top \quad i, j = 1, \dots, n$$

Information matrices

It is

$$\mathbb{E}((y_i - \mu_i)(y_j - \mu_j)) = \text{Cov}(y_i, y_j)$$

and

$$\text{Cov}(y_i, y_j) = 0 \quad \text{für } i \neq j \text{ and}$$

$$\text{Cov}(y_i, y_j) = \text{Var}(y_i) \text{ für } i = j.$$

Fisher information

$$\begin{aligned} \Rightarrow \mathbf{F}(\beta) &= \sum_{i=1}^n \left(\frac{\partial h(\eta_i)}{\partial \eta} \right)^2 \cdot \frac{\text{Var}(y_i)}{\text{Var}(y_i)^2} \cdot \mathbf{x}_i \mathbf{x}_i^\top \\ &= \sum_{i=1}^n \left(\frac{\partial h(\eta_i)}{\partial \eta} \right)^2 \cdot \frac{1}{\sigma^2} \cdot \mathbf{x}_i \mathbf{x}_i^\top \end{aligned}$$

only these
terms were
relevant
when $i = j$

$= \text{Var}(y_i)$

Information matrices

In matrix representation

$$\mathbf{y} =$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$n \times 1$

$$\boldsymbol{\mu} =$$

$$\begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix}$$

$n \times 1$

with

$$\mu_i = h(\mathbf{x}_i^\top \boldsymbol{\beta})$$

$$\eta_i$$

$$\mathbf{X} =$$

design
matrix

$$\begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix}$$

=

$$\begin{pmatrix} x_{11} & \dots & x_{1p} \\ \vdots & & \vdots \\ x_{n1} & \dots & x_{np} \end{pmatrix}$$

$n \times p$

$$\mathbf{D}(\boldsymbol{\beta}) =$$

$$\begin{pmatrix} \frac{\partial h(\eta_1)}{\partial \eta} & & 0 \\ & \ddots & \\ 0 & & \frac{\partial h(\eta_n)}{\partial \eta} \end{pmatrix}$$

(derivatives
of response
function)

$n \times n$

$$\boldsymbol{\Sigma}(\boldsymbol{\beta}) =$$

$$\begin{pmatrix} \sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & \sigma_n^2 \end{pmatrix}$$

$\sigma_i^2 \propto \text{var}(y_i)$

Estimation of GLMs

Score function

$$\begin{aligned}\Rightarrow \mathbf{s}(\beta) &= \sum_{i=1}^n \frac{\partial h(\eta_i)}{\partial \eta} \cdot \frac{y_i - \mu_i}{\sigma_i^2} \cdot \mathbf{x}_i \\ &= \mathbf{X}^\top \mathbf{D}(\beta) \Sigma(\beta)^{-1} (\mathbf{y} - \boldsymbol{\mu})\end{aligned}$$

Fisher information

$$\begin{aligned}\Rightarrow \mathbf{F}(\beta) &= \sum_{i=1}^n \left(\frac{\partial h(\eta_i)}{\partial \eta} \right)^2 \cdot \frac{1}{\sigma_i^2} \cdot \mathbf{x}_i \mathbf{x}_i^\top \\ &= \mathbf{X}^\top \mathbf{D}(\beta) \Sigma(\beta)^{-1} \mathbf{D}(\beta) \mathbf{X} \\ &= \mathbf{X}^\top \mathbf{W}(\beta) \mathbf{X}\end{aligned}$$

$\underline{\mathbf{W}}(\beta) = \underline{\mathbf{D}}(\beta) \underline{\Sigma}^{-1}(\beta) \underline{\mathbf{D}}(\beta)$ *weight matrix*

Estimation of GLMs

For canonical link

$$\Rightarrow \mathbf{s}(\beta) = \sum_{i=1}^n \frac{y_i - \mu_i}{\phi} \cdot \mathbf{x}_i = \frac{1}{\phi} \mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu})$$

$$\begin{aligned} \Rightarrow \mathbf{F}(\beta) &= \mathbb{E} \left(\mathbf{s}(\beta) \mathbf{s}(\beta)^\top \right) = \frac{1}{\phi^2} \mathbf{X}^\top \mathbb{E} \left((\mathbf{y} - \boldsymbol{\mu})(\mathbf{y} - \boldsymbol{\mu})^\top \right) \mathbf{X} \\ &= \frac{1}{\phi^2} \mathbf{X}^\top \boldsymbol{\Sigma}(\beta) \mathbf{X} \end{aligned}$$

 definition of
a covariance
matrix

Estimation of GLMs

Examples:

$$\phi = \sigma^2$$

1. Normal distribution, linear model

$$\underline{\underline{E}}(\underline{\underline{\beta}}) = \frac{1}{(\sigma^2)^n} \underline{\underline{X}}^T (\sigma^2 \underline{\underline{I}}) \underline{\underline{X}} = \frac{1}{\sigma^2} \underline{\underline{X}}^T \underline{\underline{X}}$$

$$\Leftrightarrow \underbrace{\text{cov}(\underline{\underline{\beta}}) = (\underline{\underline{X}}^T \underline{\underline{X}})^{-1} \sigma^2}_{\text{well-known}} = \underline{\underline{E}}(\underline{\underline{\beta}})^{-1}$$

2. log-linear Poisson model $\Rightarrow$ see linear models

$$\phi = 1, \quad \text{var}(y_i) = \mu_i = \pi_i$$

$$\underline{\underline{E}}(\underline{\underline{\beta}}) = \underline{\underline{X}}^T \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix} \underline{\underline{X}}$$

Estimation of GLMs

Exception with canonical link

$$\mathbf{F}(\beta) = \mathbf{F}_{\text{obs}}(\beta)$$

Proof:

Slide 6: $\frac{\partial h(\eta_i)}{\partial \eta} = \frac{\text{var}(\eta_i)}{\phi}$

$$\begin{aligned}\mathbf{s}(\beta) &= \sum_{i=1}^n \frac{y_i - \mu_i}{\phi} \cdot \mathbf{x}_i \\ \mathbf{F}_{\text{obs}}(\beta) &= -\frac{\partial \mathbf{s}(\beta)}{\partial \beta} = \sum_{i=1}^n \frac{1}{\phi} \left(\frac{\partial \mu_i}{\partial \beta} \right) \mathbf{x}_i^\top \\ &= \sum_{i=1}^n \frac{1}{\phi} \left(\frac{\partial h(\eta_i)}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta} \right) \mathbf{x}_i^\top \\ &= \sum_{i=1}^n \frac{1}{\phi} \frac{\text{Var}(y_i)}{\phi} \mathbf{x}_i \mathbf{x}_i^\top = \frac{1}{\phi^2} \mathbf{X}^\top \Sigma(\beta) \mathbf{X}\end{aligned}$$

see
slide 16

Estimation of the dispersion parameter ϕ

- By now ϕ was assumed to be known, now let it be unknown:

True value

$$\phi = \frac{\text{Var}(y_i)}{v(\mu_i)}$$

$$v(\mu_i) = b''(\eta_i) = \frac{\text{var}(y_i)}{\phi}$$

Estimated value

$$\hat{\phi} = \frac{1}{n-p} \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)},$$

with $\hat{\mu}_i = h(\mathbf{x}_i^\top \hat{\boldsymbol{\beta}})$ and $p = \text{rang}(\mathbf{X})$ (number of coefficients)

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix} \Rightarrow p+1 \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$$

Estimation of the dispersion parameter ϕ

Examples:

1. Normal distribution

$$\hat{\phi} = \hat{\sigma}^2 \Rightarrow \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{\mu}_i)^2$$
$$\left(= \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n-p-1} \sum_{i=1}^n \varepsilon_i^2 \right)$$

2. Gamma distribution

$$y \sim \text{Ga}(\nu, \alpha) \quad , \quad \nu, \alpha > 0 \quad , \quad \mu = \frac{\nu}{\alpha}$$

$$\Rightarrow \phi = \frac{1}{\nu} \quad \text{Var}(y) = \phi \cdot \mu^2 = \frac{\nu}{\alpha^2} \Rightarrow \nu(\mu) = \mu^2$$

$$\Rightarrow \hat{\phi} = \frac{1}{n-p-1} \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i^2}$$